**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# LESSON 138 · Preventing Network Attacks

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. What is the concept of SQL injection?

Q2. What does a firewall control?
A. A monitor's brightness
B. Allowed network traffic
C. All files' storage size
D. The CPU's clock speed

Q3. “HTTP is the protocol used only to send email to a mail server.” Correct this. Explain why.

Q4. A pupil logs in to a website. “HTTPS proves every website is honest.” Correct this. What does HTTPS protect?

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# I Do · Read new knowledge

### Key vocabulary

Firewall — Filters network traffic according to predefined rules.

Encryption — Makes data unreadable without the appropriate key.

Authentication — Confirms identity; MFA uses multiple factors.

Updates and backups — Patching reduces exploitable vulnerabilities; recoverable backups reduce loss after incidents.

### New learning

Firewall: Filters network traffic according to predefined rules.

Encryption: Makes data unreadable without the appropriate key.

Authentication: Confirms identity; MFA uses multiple factors.

Updates and backups: Patching reduces exploitable vulnerabilities; recoverable backups reduce loss after incidents.

### Worked example

A school wants remote access for staff while protecting student data. Recommend layered controls.

1. Require authenticated accounts and MFA.

2. Use encrypted connections to protect data in transit.

3. Filter unnecessary traffic using appropriate firewall policies.

4. Apply updates and maintain tested backups; explain different risk each control addresses.

# We Do · Practise with guidance

### Guided 1

A company uses only passwords, with no MFA. Explain an improvement.

Worked method: Require user training for phishing-resistant practice.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**





**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

## Code First · Predict → Write → Test → Improve

Change the AND condition so both checks are required before access is permitted.

```
passwordOK = true
secondFactorOK = false
if passwordOK OR secondFactorOK then
    print("Allow")
else
    print("Deny")
endif
```

iterate

### Question 1

Explain why encryption protects intercepted information. [3 marks]

### Question 2

State one role of a firewall. [2 marks]

### Question 3

Explain why software updates can improve security. [2 marks]

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

## Exit Ticket · Answer before checking

Encryption alone guarantees nobody will ever steal a password.





**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

## Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. Unsafe input changes an SQL query. An attacker may then view or change data.

Q2. B — Allowed network traffic

Q3. Transferring web resources between web clients and servers.

Q4. HTTPS encrypts data in transit. It does not prove the website or its owner is honest.

### I Do / We Do pages · Guided practice

Q1. Require user training for phishing-resistant practice.

### You Do pages · Code First

```
passwordOK = true
secondFactorOK = false
if passwordOK AND secondFactorOK then
    print("Allow")
else
    print("Deny")
endif
```

Expected result: Deny

### You Do pages · Independent questions

Q1. The interceptor sees ciphertext; without the relevant key it is not readily readable.

Q2. Allow or block traffic according to filtering rules.

Q3. Fix known vulnerabilities that might otherwise be exploited.

### Exit Ticket · Model answer

False

## Try the question again

Choose one question you got wrong. All correct? Choose the hardest.